TSM 初学者教学版 (中文) v20261016-0756

① 一句话

Taiwan Semiconductor Manufacturing Company Limited 是一家 Foundry / Advanced packaging 晶圆代工/先进封装 公司。现在股价 $422.25。我们的 Base 模型认为每股价值约 $341（-19.2%）。

② 发生了什么

最近一季财报 EPS 实际 vs 预期: [‘2026-09-30’, ‘4.6’, ‘4.52449’, ‘0.0167’]（实际、预期、超预期%）。下次财报约 2026-10-15。

③ 术语 Glossary

- DCF (Discounted Cash Flow｜现金流折现): 把未来每年能拿到的现金，按”时间成本”折回今天加总。
- FCF (Free Cash Flow｜自由现金流) = 经营现金流 − 资本开支。
- WACC (Weighted Average Cost of Capital｜加权平均资本成本): 投资人要求的年回报率，用作折现率。
- Reverse DCF (反向 DCF): 反过来问”现价要求公司增长多快”。
- Consensus (一致预期): 分析师预测的平均值。
- Falsifier (证伪指标): 一旦出现就说明论点错了的信号。

④ 生活比喻

买这家公司股票≈买一棵果树。DCF 就是估算果树未来每年结多少果(FCF)，再考虑”明年的一斤果不如今天的一斤果值钱”(WACC 折现)。

⑤ 因果箭头

Foundry / Advanced packaging 晶圆代工/先进封装 需求 → 收入增长 (g) → 自由现金流 (FCF = 收入 × FCF 率) → 折现 (÷(1+WACC)^t) → 每股价值 → 与股价比较 → 预期差

⑥ 数字例题（符号定义）

设 R_0 = 过去 12 个月收入 = 4,440.5 bn TWD；g_1 = 第一年增速 = 35.2%；m = 长期 FCF 率 = 30.0%；W = WACC = 11.4%。 第一年收入 $R_1 = R_0 \times (1+g_1)$ = 6,004.0 bn TWD。 若第 5 年 FCF 率达到 m，则第 1 年 FCF ≈ $R_1 \times m_1$（m_1 由当前 25.3% 向 30.0%

线性过渡）= 1,576.8 bn TWD； 折现到今天: $FCF_1 \div (1+W)^1$ = 1,415.6 bn TWD。十年加总 + 终值 = EV 54,784.3 bn TWD。

⑦ 事实 / 预测 / 市场预期

- 事实: 过去财务数据(第 4 节)、现价。
- 预测(我们的假设): 长期 FCF 率 30.0%、三情景。
- 市场预期: 一致预期增速 35.2%；价格隐含增速 41.4%。

⑧ 三情景

Bear $132 / Base $341 / Bull $635；概率加权 $362。

⑨ 反方观点与失效条件

反方: Geopolitical shock, US tariffs on chips, Intel/Samsung regain share｜地缘冲击、美国芯片关税、英特尔/三星夺回份额 失效条件: GM falls below long-term target band; monthly revenue y/y decelerates sharply; capex cut｜毛利率跌破长期目标区间；月营收同比骤降；Capex 下修

自我检验题

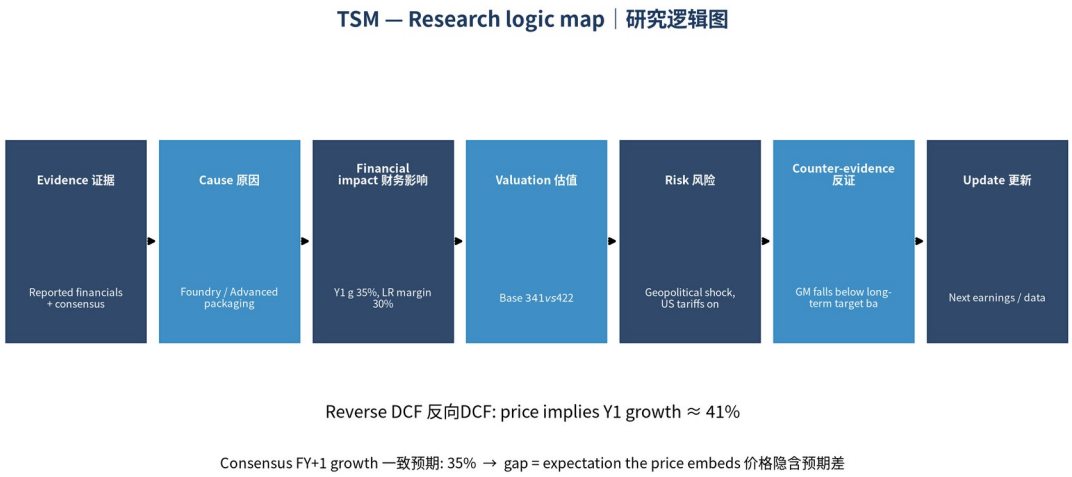
1. 如果 WACC 上升 1 个百分点，价值会升还是降？为什么？（提示：分母变大）
2. 反向 DCF 隐含增速高于一致预期意味着什么？

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① In one line

Taiwan Semiconductor Manufacturing Company Limited is a Foundry / Advanced packaging 晶圆代工/先进封装 company. Price $422.25; our Base model value ≈ $341 (-19.2%). ## ② What happened Last quarter EPS actual vs consensus: [‘2026-09-30’, ‘4.6’, ‘4.52449’, ‘0.0167’]. Next earnings ≈ 2026-10-15 (TENTATIVE). ## ③ Glossary DCF (Discounted Cash Flow｜现金流折现); FCF (Free Cash Flow｜自由现金流) = operating cash flow − capex; WACC (Weighted Average Cost of Capital｜加权平均资本成本) = the yearly return investors demand; Reverse DCF (反向 DCF) = what growth the price requires; Consensus (一致预期); Falsifier (证伪指标). ## ④ Analogy Owning the stock ≈ owning a fruit tree: DCF

estimates future fruit (FCF) and discounts it because fruit next year is worth less than fruit today. ## ⑤ Causal arrows Demand → revenue growth (g) → FCF (= revenue × FCF margin) → discounting ÷(1+WACC)^t → value/share → compare with price → expectation gap ## ⑥ Worked example (symbols) R_0 = TTM revenue = 4,440.5 bn TWD; g_1 = 35.2%; m = long-run FCF margin = 30.0%; W = 11.4%. $R_1 = R_0 \times (1+g_1)$ = 6,004.0 bn TWD; FCF_1 = 1,576.8 bn TWD; PV = $FCF_1/(1+W)$ = 1,415.6 bn TWD; 10 years + terminal → EV 54,784.3 bn TWD. ## ⑦ Fact vs forecast vs market expectation Facts: reported financials, price. Forecast: our margin/scenario assumptions. Market: consensus growth 35.2%; price-implied growth 41.4%. ## ⑧ Three scenarios Bear $132 / Base $341 / Bull $635; weighted $362. ## ⑨ Counterview & failure conditions Geopolitical shock, US tariffs on chips, Intel/Samsung regain share｜地缘冲击、美国芯片关税、英特尔/三星夺回份额 — Falsifiers: GM falls below long-term target band; monthly revenue y/y decelerates sharply; capex cut｜毛利率跌破长期目标区间；月营收同比骤降；Capex 下修 ## Self-check 1. WACC +1pp → value up or down? 2. What does price-implied growth > consensus mean?



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